

Proposed Research Design

Where does machine-learning alpha come from after trading costs? Hypotheses, architecture, data, methods, benchmarks, experiments and statistics

Machine Learning for Alpha Signal Combination and Portfolio Construction

This report is part of a pre-empirical research proposal. It distinguishes verified evidence, research inference, proposed design choices and hypotheses. No backtest results are reported; expected outcomes are hypotheses.

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Proposed Research Design: Where Does Machine-Learning Alpha Come From After Trading Costs?

Working title: "Nonlinearity, State Dependence, and Cost-Aware Learning: Decomposing the Net Value of Machine-Learning Signal Combination".

Everything in this document is a **proposed** design. Expected outcomes are **hypotheses**, not results.

1. Research Question

How much of the net-of-cost investment value of machine-learning signal combination is attributable to (i) **nonlinear** signal interactions, (ii) **state-dependent** signal weights, (iii) a **cost-aware economic training objective**, and (iv) **forecast-uncertainty** shrinkage? Do these ingredients interact? Is the learned state dependence consistent with economic theory?

2. Hypotheses (pre-registered)

ID	Hypothesis	Primary test	Grounding
H1	Nonlinearity raises gross IC and gross Sharpe relative to linear combination, but at least 50% of the gross Sharpe gain disappears net of costs	Clark-West (IC/R ²), LW2008 Sharpe difference gross vs net, factorial contrast	Gu, Kelly & Xiu (2020); Avramov, Cheng & Metzker (2023); Blitz et al. (2023)
H2	State-dependent weights add net value only when costs are internalised in training (positive State × Objective interaction)	Factorial interaction contrast, block bootstrap; SPA	Collin-Dufresne, Daniel & Saglam (2020); Garleanu & Pedersen (2013); DeMiguel, Martin-Utrera & Uppal (2024)
H3	The cost-aware objective is the largest main effect on net utility	Factorial main effects; Shapley attribution	Jensen et al. (2026); DeMiguel et al. (2020)
H4	Uncertainty shrinkage raises net Sharpe, mainly by reducing turnover	Mediation: turnover change vs gross alpha change	Liao et al. (2025); Barroso & Saxena (2022)
H5	Learned state dependence matches theory: lower momentum weight after market declines with high volatility; stronger short-leg anomaly weights after high sentiment; faster trading in liquid states	Regressions of implied weights / turnover on states	Daniel & Moskowitz (2016); Stambaugh, Yu & Yuan (2012); Collin-Dufresne, Daniel & Saglam (2020)
H6	Gross ML gains load on limits-to-arbitrage stocks; cost-aware learning shifts exposure toward liquid stocks	Characteristic tilts of portfolios	Avramov, Cheng & Metzker (2023)

Null results are informative. If H2 fails, the paper reports that conditioning adds no net value even with costs internalised. That is an important negative finding for practitioners (e.g., for the regime resampling in BlackRock Systematic (2025)).

3. Architecture

Empirical research pipeline (shaded stages = factorial ML core)

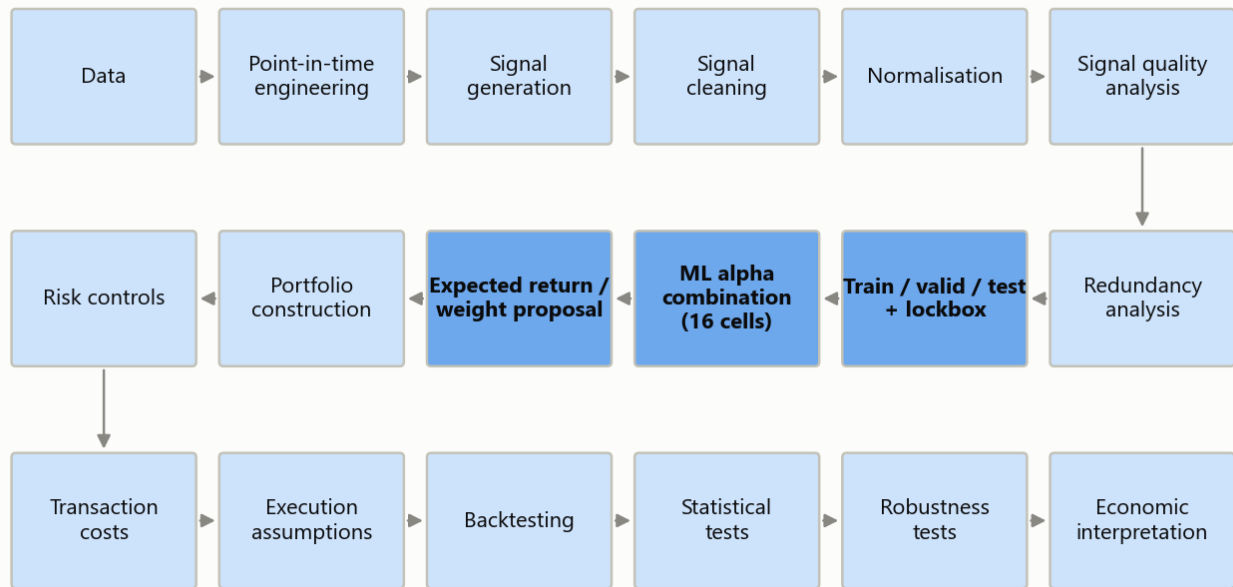


Figure 1. End-to-end empirical pipeline.

	Prediction loss no uncertainty	Prediction loss + uncertainty	Economic loss no uncertainty	Economic loss + uncertainty
Linear, static	L-S-P-0 E20	L-S-P-U E20 + E28	L-S-E-0 E24	L-S-E-U E24 + E28
Linear, state-conditional	L-C-P-0 E22	L-C-P-U E22 + E28	L-C-E-0 E26	L-C-E-U E26 + E28
Nonlinear, static	N-S-P-0 E21	N-S-P-U E21 + E28	N-S-E-0 E25	N-S-E-U E25 + E28
Nonlinear, state-conditional	N-C-P-0 E23	N-C-P-U E23 + E28	N-C-E-0 E27	N-C-E-U E27 + E28

Economic loss = net-of-cost utility. All 16 cells share signals, universe, splits, optimiser, risk and cost model.
Dark cell = full model; E29 decomposes main effects and interactions.

Figure 2. The 16-cell factorial design (experiment IDs from Workbook 4).

Point-in-time data (CRSP, Compustat, JKP/OpenAP signals, state variables, cost inputs)

- > Signal construction (~150 signals, 13 themes) + publication-date flags
- > Cross-sectional rank normalisation, missing-value indicators
- > Signal quality & redundancy diagnostics (IC, decay, turnover, clusters)
- > Walk-forward splits (expanding train, 5y validation, embargo = horizon, annual refit; 2021-2025 lockbox)
- > 16-cell factorial combination models
 - Form: Linear (L) | Nonlinear (N)
 - Conditioning: Static (S) | State-conditional (C)
 - Objective: Prediction loss (P) | Economic net-of-cost loss (E)
 - Uncertainty: None (0) | Ensemble-uncertainty shrinkage (U)
- > P-cells: standardized scores -> common alpha scaling -> common TC-aware optimiser
- > E-cells: weights learned under net utility, then projected onto the same constraint set
- > Risk controls (neutralities, position/trade limits, factor risk model)
- > Execution & costs (spread + power-law impact; AUM grid; borrow costs)
- > Backtest -> statistics (SPA/StepM/MCS, LW2008, DSR, PBO, NW alphas)
- > Factorial & Shapley attribution -> mechanism tests -> robustness & multiverse

Design principle (proposed). All cells share the signal library, universe, splits, constraint set, risk model, cost model, AUM and evaluation code. The only differences between cells are the four switched ingredients, so differences in net performance can be attributed to them.

4. Data

Component	Source	Access	Notes
Returns, prices, volume, shares, delistings	CRSP US Stock (monthly & daily) via WRDS	University subscription	Share codes 10/11; exchanges NYSE/AMEX/NASDAQ; delisting returns applied
Fundamentals	Compustat North America (annual & quarterly) via WRDS	Subscription	Lagged availability (see Section 9); CCM link table
Characteristics	JKP global characteristics (WRDS; code public) and/or Open Source AP firm-level signals (209 downloadable + 3 from CRSP)	WRDS / free download	JKP factor returns under CC BY-NC 4.0 (jkpfactors.com); OpenAP data through Dec 2024
Factor benchmarks	Ken French data library; q-factors (global-q.org); JKP theme factors	Free	FF5+UMD, q5, JKP themes
State variables	FRED (credit/term spreads, CPI, yields), Goyal predictor data, CBOE VIX (from 1990), Baker-Wurgler sentiment, Pastor-Stambaugh liquidity	Free (verify latest update dates)	All lagged one month
Trading-cost inputs	Chen-Velikov effective spreads (hf-spreads-all code, TAQ-based); EDGE OHLC estimator (Ardia, Guidotti & Kroencke, 2024); Abdi-Rinaldo (Abdi & Rinaldo, 2017); Hasbrouck Gibbs (Hasbrouck, 2009)	TAQ via WRDS; estimators from CRSP OHLC	Hierarchy: TAQ effective spread (1993+), else EDGE, else AR
Borrow costs (optional)	Markit/IHS securities finance via WRDS if licensed (JKMP code references short-selling fees from WRDS)	Institution-dependent	Fallback: institutional-ownership proxy with fee tiers

5. ML Methodology

Inputs. For stock i at month-end t : signal vector $x_{\{i,t\}}$ (about 150 cross-sectionally rank-normalised to $[-0.5, 0.5]$; missing set to 0 with 13 theme-level missing indicators); cost features $c_{\{i,t\}}$ (log effective spread, log dollar volume, Amihud illiquidity, 60-day volatility); state vector z_t (pre-specified, Section 6).

Targets. Prediction cells: cross-sectionally demeaned excess return over horizon h in $\{1, 3, 6, 12\}$ months (h chosen in validation and counted as a trial), rank-transformed in robustness. Economic cells: realised next-month net portfolio utility.

Model classes per cell.

- **L-S-P.** Ridge on x (λ grid 10^{-4} to 10^4 , 17 points). Elastic net and PLS reported as baselines.
- **L-C-P.** Ridge on $[x, xxz]$ with separate penalties for main and interaction blocks.
- **N-S-P.** LightGBM (num_leaves $\{15, 31, 63\}$; learning_rate $\{0.01, 0.05\}$; min_data_in_leaf $\{500, 2000\}$; feature_fraction $\{0.5, 0.8\}$; bagging 0.8; L2 $\{0, 10\}$; early stopping on validation, up to 2,000 trees) and an NN3 ensemble (32-16-8 ReLU, batch norm, Adam lr $1e-3$, L1 $\{1e-5, 1e-4, 1e-3\}$, early stopping patience 5, 10 seeds). The pre-specified nonlinear prediction is the equal-weight average of the LightGBM and NN ensembles; each is also reported separately.
- **N-C-P.** As N-S-P with z and xxz appended for NN and z appended for LightGBM. Gated mixture-of-experts variant: 3 NN experts, softmax gate on z only (interpretable gate).
- **L-S-E.** Linear parametric portfolio policy with costs (DeMiguel et al., 2020).
- **L-C-E.** Linear policy with coefficients linear in z (conditional PPP).
- **N-S-E.** NN policy (NN3) mapping (x, c) to weight tilts, trained on net utility (Simon, Weibels & Zimmermann, 2026; Jensen et al., 2026 Portfolio-ML as external benchmark).
- **N-C-E.** NN policy on (x, c, z) with state gate.

- **Uncertainty (U).** For P-cells, the alpha score is shrunk as $\alpha_{\text{shrunk}} = \alpha / (1 + \kappa \cdot s^2 / s_{\text{avg}}^2)$, where s is the stock's ensemble standard deviation across 10 members, s_{avg} is its cross-sectional average (seeds for NN, bagged subsamples for LightGBM or ridge) and κ in $\{0, 0.5, 1, 2, 4\}$ is chosen in validation. For E-cells, a penalty proportional to the ensemble variance of proposed weights is added. A bootstrap-SE variant (Liao et al., 2025) is used in robustness.

Economic loss (E-cells). Over training months, maximise the mean of

$$U_t = w_t' r_{t+1} - (\gamma/2) \cdot w_t' \Sigma_t w_t - TC_t(w_t - w_{+t-1}) - \eta \cdot \|w_t\|_1$$

where w_{+t-1} are drifted previous weights. TC is the cost model of Section 7 with a smooth (Huber) approximation of $|\cdot|$ for training. γ is calibrated so the ex-ante volatility of the validation-period portfolio is 10% annualised. Weights are normalised to gross exposure 2 (long 1, short 1) and projected onto the Section 7 constraint set before evaluation. Training uses truncated back-propagation through 12-month sequences so the turnover term is learned.

Alpha scaling for P-cells (common across models). score $\rightarrow$ z-score cross-sectionally $\rightarrow \alpha_{\{i,t\}} = IC_v \cdot \sigma_{\{i,t\}} \cdot \text{score}_{\{i,t\}}$ (Grinold-Kahn convention), where IC_v is the model's validation-period IC. This prevents a model from "winning" by producing larger forecast magnitudes.

6. State Variables (fixed ex ante; all observable at t)

Variable	Definition	Economic rationale
MKTVOL	log 21-day realised volatility of CRSP VW index	Momentum crashes; risk prices fall with volatility (Daniel & Moskowitz, 2016; DeMiguel, Martin-Utrera & Uppal, 2024)
BEAR	1 if CRSP VW cumulative 24-month return < 0	Momentum crash state (Daniel & Moskowitz, 2016)
ILLIQ	VW average Amihud illiquidity, detrended (log deviation from 12-month average)	Liquidity regimes (Collin-Dufresne, Daniel & Saglam, 2020; Amihud, 2002)
SENT	Baker-Wurgler sentiment (orthogonalised version), last available	Short-leg anomaly strength (Stambaugh, Yu & Yuan, 2012; Baker & Wurgler, 2006)
CREDIT	BAA – AAA yield spread	Distress / funding
TERM	10y – 3m Treasury	Business cycle
INFL	CPI YoY inflation (lagged for release)	Inflation regime
DRATE	12-month change in 10y yield	Rate regime
FMOM_k	Trailing 12-month return of theme-k factor (13 values)	Factor momentum (Ehsani & Linnainmaa, 2022)
DISP	Cross-sectional dispersion of composite signal	Opportunity set

State variables are standardised with expanding-window means and standard deviations. HMM-smoothed regime probabilities are prohibited (look-ahead). Placebo tests replace z with (i) block-shuffled z and (ii) random-walk series.

7. Portfolio Methodology

Optimiser (P-cells and projection step for E-cells).

$$\max_w \alpha_t' w - (\gamma/2) \cdot w' \Sigma_t w - \sum_i TC_i(|w_i - w_{+i,t-1}|)$$

subject to:

- $\sum_i w_i = 0$ (dollar neutral) and $\sum_i |w_i| \leq 2$
- $|\beta_t' w| \leq 0.05$ (beta from the risk model)
- $|\sum_{\{i \text{ in } g\}} w_i| \leq 0.02$ for each Fama-French 49 industry g
- $|w_i| \leq \min(1\%, 10 \times \text{participation cap})$, with trade size $|\Delta w_i| \cdot \text{AUM} \leq 5\%$ of 20-day ADV_i
- Long-only benchmark-relative variant in robustness: $w \geq 0$, $\sum w = 1$, tracking error $\leq 4\%$

Solved with CVXPY (MOSEK or Clarabel); failures are logged and the portfolio is held.

Risk model. Structural factor model $\Sigma = B F B' + D$. B is market beta, 13 JKP theme exposures (cross-sectional z-scores) and FF49 industry dummies. Factor returns come from daily cross-sectional WLS ($\sqrt{\text{market-cap}}$ weights). F is EWMA with a 126-day half-life and Newey-West adjustment. D is EWMA idiosyncratic variance with Bayesian shrinkage toward size-decile means. Robustness: a 20-factor PCA statistical model and a Ledoit-Wolf-shrunk sample covariance on the top-500 universe (Ledoit & Wolf, 2017).

Transaction-cost model (per stock, per trade of dollar size Q).

$$TC_i(Q) = (s_i/2) \cdot |Q| + k \cdot \sigma_i \cdot |Q|^{\{1.5\}} / ADV_i^{\{0.5\}} + \text{commission} \cdot |Q|$$

- s_i : effective spread (TAQ-based where available; EDGE or AR estimator otherwise).
- σ_i : daily volatility; ADV_i : 20-day average dollar volume.
- $k = 1$ baseline, with $0.5\times$, $2\times$ and $3\times$ sensitivity. The $3/2$ total-cost power corresponds to square-root impact; Almgren et al. (2005) report a concave impact exponent near $3/5$ [verify], and the functional form is a sensitivity dimension.
- commission = 1 bp.
- Short positions pay 25 bps per year (general collateral). Stocks in the bottom institutional-ownership decile pay 5% per year, or Markit fees where licensed.
- AUM baseline \$1bn (Dec-2020 dollars, scaled by total CRSP market capitalisation), grid {\$0.1bn, \$1bn, \$10bn}.

Bounds: live institutional costs may be lower (Frazzini, Israel & Moskowitz, 2018); fund-implied costs may be higher (Patton & Weller, 2020).

Execution assumption. Signals are formed with information available at month-end t . Trades are assumed executed at the close of the first trading day of month $t+1$ (one-day implementation lag), with returns accrued from that price. A same-day-close variant is reported only as a sensitivity.

8. Benchmarks

Group	Benchmarks
Naive	1/N universe; VW market; EW z-score composite; theme-EW composite; IC-weighted (rolling 60m); ICIR-weighted; inverse-volatility of signal portfolios
Linear statistical	Lewellen FM rolling slopes (Lewellen, 2015); ridge; lasso; elastic net; PLS (Light, Maslov & Rytchkov, 2017)
Portfolio-level	KNS shrunk SDF over signal portfolios (Kozak, Nagel & Santosh, 2020); NRRZ-style PLS factor timing (Neuhierl et al., 2024)
Nonlinear ML	Random forest; LightGBM; GKX NN3 (Gu, Kelly & Xiu, 2020); IPCA expected returns (Kelly, Pruitt & Su, 2019)
Economic objective	DMNU linear PPP with TC (DeMiguel et al., 2020); JKMP Portfolio-ML using authors' public code (Jensen et al., 2026); deep PPP (Simon, Weibels & Zimmermann, 2026) if code obtainable
Cost mitigation	NMV buy/hold banding applied to the EW composite (Novy-Marx & Velikov, 2016)
Factor models (alpha)	FF5 + UMD; q5; JKP 13 themes; DNMV net-of-cost model comparison logic (Detzel, Novy-Marx & Velikov, 2023)

Every benchmark receives the **same** tuning budget per refit as the corresponding factorial cell, and all hyperparameter grids are published.

9. Validation Framework

- Walk-forward.** Initial training Jan 1972 to Dec 1989; validation Jan 1990 to Dec 1994; first test year 1995. Each January: expanding training window, rolling 5-year validation immediately before the test year, embargo gap equal to the target horizon h between training/validation and validation/test (labels overlapping the next split are purged) (Lopez de Prado, 2018). Refit annually.
- Lockbox.** Jan 2021 to Dec 2025 remains untouched until all design choices are frozen and pre-registered. It is opened once.
- Rolling-window robustness.** A 20-year rolling training window.

- **No k-fold shuffling** of time. Purged grouped cross-validation is used only inside the training window for early-stopping diagnostics.
- **Point-in-time rules.** Annual Compustat items are used ≥ 4 months after fiscal year-end; quarterly items after RDQ, or ≥ 4 months after quarter-end if RDQ is missing. Market equity is at t . Signals whose original publication year is after t are flagged, and a gated library is used in robustness (McLean & Pontiff, 2016). The CRSP-Compustat link is valid at t . Survivorship: the universe is formed at t from all securities alive at t . Delisting returns come from CRSP; missing performance-delisting returns are set to -30% (Shumway, 1997; verify convention).

10. Evaluation Metrics

Group	Metrics
Prediction	Pooled OOS R^2 vs zero (Gu, Kelly & Xiu, 2020); MSE; IC, rank IC, ICIR, IC t-stat (NW); top-minus-bottom decile spread (VW)
Portfolio (gross & net)	Annualised mean, volatility, Sharpe, Sortino, max drawdown, Calmar, skewness, kurtosis, CVaR(5%)
Implementation	One-way monthly turnover; average holding period; cost drag (bps/month); share of costs from spread vs impact; short-leg share of returns; microcap/illiquid exposure
Economic	Certainty-equivalent return; net alpha vs FF5+UMD, q5, JKP themes (NW t); implementable efficient frontier (Jensen et al., 2026); break-even AUM; break-even cost multiplier
Conversion rates (novel reporting)	Net Sharpe per unit IC; cost drag per unit turnover; transfer coefficient (correlation of scores with implemented weights) (Clarke, de Silva & Thorley, 2002)

11. Statistical Framework

1. **Mean returns and alphas.** Newey-West HAC t-statistics (12 lags) (Newey & West, 1987).
2. **Sharpe differences.** Ledoit-Wolf studentised circular block bootstrap (block lengths 4, 6, 12; 5,000 reps) (Ledoit & Wolf, 2008); Lo's serial-correlation adjustment for reporting (Lo, 2002).
3. **Forecast comparisons.** Clark-West for nested models (Clark & West, 2007); Diebold-Mariano on cross-sectional average errors (Diebold & Mariano, 1995).
4. **Data snooping.** Every configuration is logged (MLflow). Hansen SPA versus the strongest benchmark (stationary bootstrap, mean block 6, 10,000 reps) (Hansen, 2005; Politis & Romano, 1994); White Reality Check (White, 2000); Romano-Wolf StepM to identify which cells beat the benchmark (Romano & Wolf, 2005); Model Confidence Set at 10% (Hansen, Lunde & Nason, 2011).
5. **Backtest overfitting.** Deflated Sharpe ratio using the logged number of trials and their correlation (Bailey & Lopez de Prado, 2014); probability of backtest overfitting via CSCV with 16 blocks (Bailey et al., 2017); Harvey-Liu haircut Sharpe (Harvey & Liu, 2015).
6. **Factorial attribution.** Monthly net returns of the 16 cells regressed on ± 1 -coded factors and all interactions (orthogonal contrasts). Shapley decomposition of net Sharpe and CE across the four ingredients. Confidence intervals by month-block bootstrap, which preserves cross-cell dependence.
7. **Stability.** Bai-Perron structural breaks in rolling cell differences (Bai & Perron, 1998); subperiod tests.
8. **Multiple signal/theme claims.** Benjamini-Hochberg FDR (Benjamini & Hochberg, 1995); $t > 3$ hurdle for any new-signal-type claim (Harvey, Liu & Zhu, 2016).
9. **Non-standard errors.** Distribution of main-effect estimates across a pre-declared multiverse (window type, universe, horizon, target transform, cost estimator, risk model) (Menkveld et al., 2024; Chen, Hanauer & Kalsbach, 2024).

12. Robustness Framework

Subperiods (1995-2000, 2001-2008, 2009-2019, 2020-2025; pre/post decimalisation). Universes (all stocks incl. microcaps; ex-microcaps (main); NYSE top 500). Weighting and constraints (long-only; no industry constraint). Horizons (1, 3, 6, 12m) and quarterly rebalancing. Cost multipliers 0.5-3× and alternative spread estimators. AUM grid. Risk models (structural, PCA, LW shrinkage). Rolling vs expanding windows. Publication-gated signal library. Placebo states. NN seeds (≥ 10) and ± 1 grid-step perturbations. International out-of-sample (JKP developed ex-US). Alternative uncertainty estimators (bootstrap SE, MC dropout, conformal).

13. Economic Interpretation Plan

- **Economic feature importance.** Change in net utility when a theme is removed and the model refit (Jensen et al., 2026). TreeSHAP and deep SHAP by theme for P-cells (Lundberg & Lee, 2017). ALE plots, given correlated signals (Apley & Zhu, 2020).
- **Interaction analysis.** Friedman H-statistics for the top theme pairs; comparison of gross vs net importance.
- **State dependence.** Implied signal-theme weights (cross-sectional regression of implemented weights on theme scores) regressed on z_t , with tests of H_5 .
- **Portfolio attribution.** Returns decomposed into theme exposures, industry, market and residual; cost attribution by theme.
- **Limits to arbitrage.** Portfolio tilts on idiosyncratic volatility, illiquidity, size and distress by cell (H_6) (Avramov, Cheng & Metzker, 2023).
- **Alignment.** Ex-ante vs ex-post risk bias statistics by cell (Ceria, Saxena & Stubbs, 2012).

14. Contribution Statement (draft)

Existing literature shows that machine-learning combinations of firm characteristics improve return prediction (Gu, Kelly & Xiu, 2020), that the resulting portfolios are fragile to economic restrictions and trading costs (Avramov, Cheng & Metzker, 2023; Blitz et al., 2023), and that learning portfolio weights under a cost-aware objective substantially improves the implementable efficient frontier (Jensen et al., 2026; DeMiguel et al., 2020; Simon, Weibels & Zimmermann, 2026).

However, these studies compare bundled pipelines. They largely assume that the sources of ML's value (flexible functional form, time-varying signal efficacy, the training objective and forecast uncertainty) can be assessed one at a time, on different data, costs and evaluation criteria.

We propose a controlled factorial design that switches each ingredient on and off within one signal library, universe, optimiser, cost model and inference framework, and we attribute net-of-cost performance to the ingredients and their interactions with data-snooping-robust tests.

This allows us to test whether nonlinearity creates value beyond its trading costs, whether state-dependent signal weights help only when costs are internalised, and whether uncertainty shrinkage works through turnover.

We demonstrate [results to be determined] across US equities from 1995 to 2025, including an untouched 2021-2025 holdout, three AUM levels, cost-model perturbations and developed international markets.

The results imply [to be determined; stated only after the lockbox is opened].

What we explicitly do not claim. We are not the first to use ML for signal combination, the first to include transaction costs, the first to condition on market states, or the first to quantify design-choice dispersion (see the closest-prior-art tracker).

15. Comparison Tables Against the Closest Papers

15.1 Ten closest papers

Paper	What it does	What we use	What we improve	Why ours is different
Jensen et al. (2026)	Cost-aware ML weight learning; implementable frontier	Frontier evaluation; Portfolio-ML as benchmark; public code	Isolate the objective effect from nonlinearity, conditioning and uncertainty	Attribution and interaction tests, not a single pipeline
Simon, Weibels & Zimmermann (2026)	NN parametric portfolio policies vs linear	Nonlinear economic-loss cell	Common cost model/optimiser; conditioning; snooping control	Decomposition and mechanisms
DeMiguel et al. (2020)	Linear PPP with TC; netting	Linear economic-loss cell	Nonlinear and conditional extensions	Measures netting benefits of nonlinear/conditional combination
DeMiguel, Martin-Utrera & Uppal (2024)	Conditional multifactor weights net of costs	Conditioning-with-costs logic	Stock-level, many states, nonlinear	Tests the conditioning × cost interaction at scale
Gu, Kelly & Xiu (2020)	ML prediction horse race	Model classes and OOS R^2 design	Net-of-cost evaluation in a common optimiser	Economic rather than statistical attribution
Avramov, Cheng & Metzker (2023)	ML under economic restrictions	Universe restrictions; limits-to-arbitrage tests	Constructive design	Explains which ingredient causes fragility
Neuhierl et al. (2024)	High-dimensional factor timing	State variables; PLS timing benchmark	Stock-level implementation with netting and costs	Timing information inside cost-aware combination
Chen, Hanauer & Kalsbach (2024)	ML design choices; non-standard errors	Multiverse reporting	Economic ingredients; net of costs; common optimiser	Attribution of net value, not dispersion
Liao et al. (2025)	NN forecast uncertainty; uncertainty-averse portfolios	Bootstrap SE variant	Cost interaction; turnover channel	Uncertainty as one factor in the factorial design
Blitz et al. (2023)	Horizon and net ML alpha	Horizon grid	Horizon chosen in validation and counted as trials	Horizon as a controlled design factor

15.2 Five strongest methodological precedents

Paper	What it does	What we use	What we improve	Why ours is different
Hansen (2005)	SPA test	Snooping control over cells	Applied to factorial cells	Cell-level inference
Romano & Wolf (2005)	StepM	Identify cells beating benchmark	-	-
Bailey et al. (2017)	PBO via CSCV	Overfitting probability of chosen configuration	Report for full trial log	Transparency
Menkveld et al. (2024)	Non-standard errors	Multiverse design	Economic ingredients	Attribution
Ledoit & Wolf (2008)	Robust Sharpe tests	Pairwise Sharpe inference	Gross vs net	Conversion rates

15.3 Five strongest portfolio-construction precedents

Paper	What it does	What we use	What we improve	Why ours is different
Garleanu & Pedersen (2013)	Dynamic trading with costs; aim portfolio	Partial adjustment logic	Learned signal weights	Empirical ML setting
Collin-Dufresne, Daniel & Saglam (2020)	Liquidity-regime allocation with costs	H5 trading-speed prediction	Empirical test at stock level	Theory test
Brandt, Santa-Clara & Valkanov (2009)	Parametric portfolio policies	Economic-loss parameterisation	Nonlinear, conditional, cost-aware	Nested cells
Ledoit & Wolf (2017)	Nonlinear shrinkage covariance	Robustness risk model	-	-
Novy-Marx & Velikov (2016)	Anomaly costs and mitigation	Banding benchmark; spread measures	Optimised mitigation	Optimiser vs heuristic

15.4 Design v2: what changed after the 2025-2026 frontier search (September 2026)

A targeted search of what these journals have accepted or circulated in 2025-2026 produced four changes. Each is a **level on an existing axis** or a **credibility instrument**; nothing was added as decoration, because every extra arm weakens the deflated Sharpe ratio of the eventual winner.

Change	What it is	Why it is now necessary
Form ladder A2-A4	Ridgeless random Fourier features on a complexity grid ($c = P/T$), a sparse basis-pursuit arm on the same feature map, and a cross-sectional attention block	The complexity debate moved on: Afsharhajari & Li (2026) shows complexity pays through sparse discovery while dense ridgeless plateaus, and Kelly et al. (2025) shows it pays only when the factor spectrum is diffuse. Attention is the current state of the art for cross-asset combination (Kelly et al., 2025; Lai, 2025)
Uncertainty ladder D2-D3	Time-ordered split conformal intervals with adaptive recalibration, feeding an ellipsoidal or Wasserstein robust objective	Ensemble dispersion has no coverage guarantee; conformal does (Kato, 2024), and connecting it to the decision is the economically interesting claim
Economic-objective controls	Output clipping, partial adjustment and a reported prediction-inflation diagnostic	Wang & Hasuikie (2026) shows decision-focused learning inflates signals and turnover, which is exactly the pathology our objective axis tests
Credibility layer	Zero-predictability synthetic null, microstructure placebo, multiplicity-adjusted inflation gap, spectrum diagnostic, limits-to-learning caveat	[verify authors] (2026) shows adaptive specification search produces significant backtests under a no-predictability null; Chen, Kelly & Malamud (2025) shows empirical fit understates true predictability

New closest prior art. Zhang (2025) (October 2025) performs friction-aware, regime-conditioned policy optimisation with costs inside the reward and a regime x cost evaluation grid. It is the nearest neighbour to gap G2 and must be cited in the introduction. The differentiation is structural: it proposes a single reinforcement-learning method in a small asset space, whereas this paper identifies *which ingredient* produces net-of-cost value in a ~150-signal stock cross-section under a nested factorial design with data-snooping-robust inference.

Deliberately excluded, with reasons recorded so referees see the choice was made rather than missed: deep reinforcement-learning agents (occupied by Zhang (2025); replication risk), graph neural networks on licensed relationship data (point-in-time integrity), LLM-generated signals (training data covers our sample, so look-ahead is unresolvable before a strict post-cutoff window), and diffusion models for return distributions (no finance precedent; adds trials without touching an axis).

15.5 Five strongest ML-finance precedents

Paper	What it does	What we use	What we improve	Why ours is different
Gu, Kelly & Xiu (2020)	ML prediction benchmark	NN3/GBDT designs	Net evaluation	Attribution
Chen, Pelger & Zhu (2024)	State-dependent deep SDF	Macro-state conditioning idea	Costs; interpretable fixed states	Implementable conditioning
Kelly, Pruitt & Su (2019)	IPCA	Benchmark expected returns	-	-
Azevedo, Hoegner & Velikov (2023)	ML strategies net of costs	Net evaluation conventions	Cost-aware learning	Attribution
Cong et al. (2021)	RL/transformer direct construction	Direct-objective idea	Controlled comparison under strict costs	Replicability focus

Data Strategy

1. Principles

1. **Point-in-time or nothing.** Every variable carries the timestamp at which it was knowable.
2. **University-obtainable first.** The main paper uses WRDS-standard data plus free, documented datasets. Costly alternative data are reserved for extensions.
3. **Reproducibility.** Code must run end-to-end from raw WRDS extracts and public downloads, consistent with the JF, RFS and Management Science data/code policies (verified; see Journal Strategy).
4. **Bias register.** Every source has a bias assessment (below). Items marked **[verify]** must be confirmed against vendor documentation when data are pulled.

2. Source Assessment

Category	Source	Availability / access	Cost to university	Historical depth	Survivorship bias	Point-in-time	Look-ahead risk	Corporate actions	Licensing
Prices, returns, volume, shares	CRSP US Stock (WRDS)	Standard subscription	Institutional licence	Monthly from 1925/26; daily history [verify range]	Low if delisted securities kept; apply delisting returns	Yes (market data)	Low; beware using end-of-month data for same-month trading	CRSP adjustment factors (cfacpr, cfacshr); distributions in returns	No redistribution of raw data
Fundamentals	Compustat North America (WRDS)	Standard	Institutional	1950s; reliable coverage from ~1963 [verify]	Backfill bias in early years; require ≥2 years of history	Partial: restated data overwrite originals unless Snapshot/PIT product used	High if accounting lags ignored	Via CCM link table	No redistribution
Point-in-time fundamentals	Compustat Snapshot (WRDS)	Additional subscription [verify availability]	Additional	Shorter history [verify]	Low	Yes	Low	-	No redistribution
Link table	CRSP/Compustat Merged (CCM)	Standard	Included	Full	-	Use link dates	Medium if link dates ignored	-	-
Characteristics	JKP global factor/characteristics (code public; data on WRDS; factor returns at jkp factors.com, CC BY-NC 4.0, through Dec 2025)	WRDS for stock-level; free factor returns	Free/institutional	Long US history; 93 countries	Inherits CRSP/Compustat	Construction lags per JKP code [verify]	Low if code used as is	Handled	CC BY-NC 4.0 (non-commercial) for downloadable data
Signals	Open Source Asset Pricing (Chen-Zimmermann)	Free download: 209 firm-level signals (+3 from CRSP), monthly/daily portfolio returns; data through Dec 2024	Free	Varies by signal	Inherits sources	Signal code lags per original papers	Low-medium: verify each signal's timing	Handled	Cite CZ2022; CRSP-based signals excluded for licensing
Analyst data	I/B/E/S (WRDS)	Standard	Institutional	US detail from 1970s/80s [verify]	Low	Use announcement/activation dates	Medium (revision timestamp)	Split-adjusted vs unadjusted files	No redistribution
Institutional holdings	Thomson/Refinitiv 13F (WRDS)	Standard	Institutional	1980+	Low	Filing-date lag (45 days)	Medium if report date used instead of filing date	-	No redistribution
Short interest	Compustat short interest supplemental (WRDS)	Standard [verify coverage by exchange/year]	Institutional	1970s+ for NYSE/AMEX [verify]	Low	Publication lag	Medium	-	-
Securities lending / borrow fees	IHS Markit Securities Finance (WRDS)	Additional licence	High	~2000s+ [verify]	Low	Daily	Low	-	Restricted
Intraday / spreads	TAQ (WRDS)	Standard at many universities	Institutional	Monthly TAQ from 1993; daily TAQ from 2003 [verify]	-	Yes	Low	-	Restricted

Category	Source	Availability / access	Cost to university	Historical depth	Survivorship bias	Point-in-time	Look-ahead risk	Corporate actions	Licensing
Spread estimators	Chen-Velikov hf-spreads-all code (TAQ-based effective spreads); EDGE (Ardia, Guidotti & Kroencke, 2024) via R package bidask; Abdi-Rinaldo (Abdi & Rinaldo, 2017); Corwin-Schultz (Corwin & Schultz, 2012); Hasbrouck Gibbs (Hasbrouck, 2009)	Code public; inputs CRSP/TAQ	Free code	CRSP OHLC history [verify high/low coverage pre-1993]	-	Estimated from past prices only	Low if rolling windows end at t	-	Code licences on GitHub/CRAI
Options	OptionMetrics IvyDB (WRDS)	Additional licence	Medium-high	1996+	Low	Yes	Low	-	Restricted
Macro / states	FRED (St. Louis Fed); Goyal predictor data; CBOE VIX; Baker-Wurgler sentiment; Pastor-Stambaugh liquidity	Free	Free	1950s-present (VIX from 1990)	-	Beware revisions (use ALFRED vintages for revised macro series)	Medium for revised macro data (CPI less affected)	-	Free with attribution
Factor returns	Ken French data library; global-q.org q-factors; JKP themes	Free	Free	1926+ (FF3)	-	Yes	Low	-	Free with attribution
News / sentiment	RavenPack (WRDS)	Additional licence	High	2000+	Low	Timestamped	Medium (entity mapping)	-	Restricted
Text	SEC EDGAR filings; Loughran-McDonald dictionaries	Free	Free	1993/1996+	Low	Filing timestamps	Low	-	Free
ESG	MSCI/Refinitiv/Sustainalytics	Licence	High	2000s+ (short)	High (coverage expansion)	Often not PIT; rating revisions	High	-	Restricted; excluded from main study
Supply chain	Compustat Segments customer data (free with Compustat); FactSet Revere (licence)	Mixed	Medium-high	1970s+ segments	Medium	Filing lags	Medium	-	Restricted
International equities	Compustat Global / Datastream; JKP international characteristics	Licence / WRDS	Institutional	1990s+ for many markets	Medium (coverage changes)	JKP code lags	Medium	Vendor-specific	Restricted
LLM / embeddings	Public LLMs	Free/paid	Varies	-	-	Training-data look-ahead	Very high (model trained on future text)	-	Terms of use; excluded from main study

3. Minimum Viable Dataset for the Main Paper

1. CRSP monthly and daily (1963-2025), including delisting file.
2. Compustat annual and quarterly with CCM links.
3. JKP characteristics (WRDS) **or** OpenAP firm-level signals plus CRSP. Recommended: JKP as the primary library for its theme structure and global extension; OpenAP for signal replication checks and publication dates.
4. TAQ-based effective spreads (1993+) plus EDGE/AR estimators from CRSP OHLC for the full history.
5. Ken French factors, q-factors and JKP theme factors.
6. FRED, Goyal, VIX, Baker-Wurgler and Pastor-Stambaugh state variables.

Optional: IBES, 13F and short interest (for crowding and limits-to-arbitrage mechanism tests); Markit borrow fees; JKP international for robustness.

4. Point-in-Time Engineering Rules

Issue	Rule
Accounting availability	Annual items usable ≥ 4 months after fiscal year-end (follow JKP code conventions; verify); quarterly items at RDQ if available, else ≥ 4 months after quarter-end
Restatements	Main: Compustat as provided with lags. Robustness: Compustat Snapshot if available
Market equity	Month-end t price $\times$ shares outstanding (CRSP)
Delisting	CRSP dlret; missing performance-related delisting returns set to -30% (NYSE/AMEX convention per Shumway 1997; verify NASDAQ convention)
Survivorship	Universe built at t from all securities trading at t
Corporate actions	CRSP total returns; adjustment factors for prices/shares in signals
Microcaps	Main universe excludes stocks below NYSE 20th size percentile and price $< \$5$ at t
Signal publication	Flag signals by original publication year (OpenAP metadata); gated library robustness
Macro revisions	Use ALFRED vintages for revised series (industrial production, GDP) if used; prefer market-based states
Regime labels	No smoothed (full-sample) regime probabilities; only filtered or threshold states
Standardisation	Cross-sectional ranks at t only; time-series standardisation with expanding windows
Hyperparameters	Chosen on validation windows strictly before test year
Missing data	Median/zero imputation after ranking plus missing indicators; robustness with cross-sectional/time-series imputation (Bryzgalova et al., 2025; Freyberger et al., 2025)

5. Data Quality Tests (acceptance criteria for Phase 2)

- Rebuilt market, size and value factors correlate > 0.95 with Ken French series (monthly, 1972-2025).
- Signal long-short t-statistics reproduce OpenAP/JKP values: regression slope and R^2 comparable to Chen & Zimmermann (2022).
- No return observations after a delisting date; no duplicate PERMNO-months.
- Accounting variables never used before their availability date (automated unit tests).
- Spread estimates: cross-sectional correlation with TAQ effective spreads reported for the 1993+ overlap.

Experimental Framework and Implementation Specification

This document is written to be handed directly to a quant developer. Parameter values are the **proposed baseline**. Any change after the lockbox specification freeze (end of Phase 7) must be logged as a deviation.

1. End-to-End Research Pipeline

#	Stage	What happens	Key controls	Output artefact
1	Data	Pull CRSP, Compustat, CCM, TAQ spreads, factors, states	Versioned raw extracts with pull dates	data/raw/* + manifest
2	Point-in-time engineering	Lag accounting data, link identifiers at t, apply delisting returns, build month-end panels	Automated look-ahead unit tests	panel_pit.parquet
3	Feature / signal generation	Compute ~150 JKP/OpenAP signals; publication-year flags	Replication tests vs published t-stats	signals.parquet
4	Signal cleaning	Remove signals with <50% average coverage; handle outliers via ranks	Coverage report by year	signal_qc.csv
5	Normalisation	Cross-sectional rank to [-0.5, 0.5]; missing -> 0 + theme missing indicators	Only information at t	X_t arrays
6	Signal quality analysis	IC, rank IC, ICIR by horizon; IC half-life; turnover of signal portfolios	NW t-stats	E02 report
7	Redundancy analysis	Rank-correlation and return-correlation clustering; effective number of signals	Bootstrap stability	E03 report
8	Train/validation/test	Expanding train from 1972, 5y validation, embargo = h, annual refit, 1995-2020 test, 2021-2025 lockbox	Split calendar file checked into repo	splits.yaml
9	ML alpha combination	16 factorial cells + benchmarks	Identical tuning budgets; all trials logged	Model registry
10	Expected return / ranking	P-cells -> standardized scores -> IC-scaled alpha; E-cells -> weight proposals	Common scaling rule	alpha_t, w_prop_t
11	Portfolio construction	TC-aware MVO with constraints; E-cell projection	Solver status logs	weights_t
12	Risk controls	Dollar/beta/industry neutrality; position, trade and participation limits	Constraint-violation checks	Risk report
13	Transaction costs	Spread + power-law impact + commission + borrow	Cost multipliers; AUM grid	Cost ledger
14	Execution assumptions	Trade at close of first trading day of t+1; returns from execution price	One-day lag enforced in code	Execution log
15	Backtesting	Monthly P&L gross/net; drifted weights	Reconciliation: Σ contributions = portfolio return	returns_cells.parquet
16	Statistical tests	NW, LW2008, CW, DM, SPA, StepM, MCS, DSR, PBO	Trial log feeds DSR/PBO	Inference tables
17	Robustness	Subperiods, universes, costs, AUM, horizons, risk models, placebo, gating, seeds, international	Pre-declared multiverse	Robustness appendix
18	Economic interpretation	Economic feature importance, SHAP/ALE, implied state-dependent weights, limits-to-arbitrage tilts	Tie each result to hypothesis H1-H6	Interpretation section

2. Specification Summary (configuration)

```

universe:
  source: CRSP
  share_codes: [10, 11]
  exchanges: [NYSE, AMEX, NASDAQ]
  main_filter: {min_price: 5, exclude_below_nyse_size_pct: 20}
  robustness_universes: [all_stocks, nyse_top500]
frequency: {signals: monthly, rebalancing: monthly, risk_and_costs: daily}
periods:
  initial_train: [1972-01, 1989-12]
  initial_validation: [1990-01, 1994-12]
  walk_forward_test: [1995-01, 2020-12]
  lockbox: [2021-01, 2025-12]
  refit: annual_january
  train_window: expanding # robustness: rolling_240m
  validation_window: 60m_rolling
  embargo_months: horizon_h
signals:
  library: JKP_characteristics # cross-check: OpenAP
  themes: 13
  normalisation: cross_sectional_rank[-0.5,0.5]
  missing: zero_plus_theme_indicators
  publication_gating: robustness_only
states: [MKTVOL, BEAR, ILLIQ, SENT, CREDIT, TERM, INFL, DRATE, FMOM_theme, DISP]
targets: {prediction_horizons: [1, 3, 6, 12], choose_in_validation: true}
factorial:
  form: [linear, nonlinear]
  conditioning: [static, state]
  objective: [prediction, economic]
  uncertainty: [none, ensemble_shrinkage]
models:
  ridge: {lambda_grid: logspace(-4, 4, 17)}
  lightgbm: {num_leaves: [15, 31, 63], learning_rate: [0.01, 0.05], min_data_in_leaf: [500, 2000],
    feature_fraction: [0.5, 0.8], bagging_fraction: 0.8, lambda_l2: [0, 10],
    max_trees: 2000, early_stopping: validation}
  nn3: {layers: [32, 16, 8], activation: relu, batch_norm: true, optimizer: adam, lr: 0.001,
    l1: [1e-5, 1e-4, 1e-3], early_stopping_patience: 5, ensemble_seeds: 10}
  moe: {experts: 3, gate_inputs: states_only}
  economic_policy: {net: nn3, bptt_months: 12, gamma: calibrate_to_10pct_vol, huber_delta: 1e-4}
  uncertainty: {members: 10, kappa_grid: [0, 0.5, 1, 2, 4]}
portfolio:
  objective: mean_variance_minus_tc
  constraints: {dollar_neutral: true, gross_max: 2.0, beta_abs_max: 0.05,
    industry_ff49_abs_max: 0.02, weight_abs_max: 0.01, adv_participation_max: 0.05}
  long_only_variant: {tracking_error_max: 0.04}
  solver: [MOSEK, CLARABEL]
risk_model:
  type: structural
  factors: [market_beta, jkp_13_themes, ff49_industries]
  factor_cov: {ewma_halflife_days: 126, newey_west: true}
  specific: {ewma_halflife_days: 126, bayesian_shrink: size_deciles}
  robustness: [pca_20, ledoit_wolf_top500]
costs:
  spread: {primary: taq_effective_spread, fallback: [edge_ohlcv, abdi_ranaldo]}
  impact: {form: "k * sigma_daily * |Q|^{1.5} / ADV^{0.5}", k: 1.0, sensitivity: [0.5, 2.0, 3.0]}
  commission_bps: 1
  borrow: {general_collateral_bps_pa: 25, hard_to_borrow_pa: 0.05, htb_proxy: bottom_decile_inst_ownership}
  aum_usd_2020: [1.0e8, 1.0e9, 1.0e10]
  aum_scaling: total_crsp_market_cap
execution: {lag_days: 1, price: close}
inference:
  nw_lags: 12
  lw2008_blocks: [4, 6, 12]
  spa: {bootstrap: stationary, mean_block: 6, reps: 10000}
  mcs_alpha: 0.10
  pbo_cscv_blocks: 16
  bootstrap_reps: 5000
logging: {tracker: mlflow, log_every_trial: true}

```

3. Experiment Register

The experiment register (IDs E00-E64: hypothesis, model, dataset, benchmark, metrics, statistical test, expected outcome, status) is in Workbook 4. Summary by block:

Block	Experiments	Purpose
Data integrity	E00	PIT audit and factor replication
Signal library	E01-E03	Replication, quality/decay, redundancy
Baselines	E10-E14	Naive, linear, portfolio-level, published ML, linear cost-aware

Block	Experiments	Purpose
Factorial core	E20-E29	16 cells + decomposition
Portfolio & costs	E30-E34	Frontier, cost sensitivity, capacity, constraints, heuristics
Inference	E40-E43	SPA/StepM/MCS, DSR/PBO, alphas, multiverse
Robustness	E50-E56	Subperiods, universes, horizons, placebo, gating, international, seeds
Interpretation	E60-E64	Economic importance, state-dependent weights, limits to arbitrage, trading speed, alignment

4. Bias Controls Matrix

Bias	Where it enters	Control in this design
Look-ahead	Accounting lags; regime labels; standardisation; hyperparameters	PIT rules; filtered states only; cross-sectional ranks; validation-only tuning; unit tests
Survivorship	Universe construction	Universe at t from all live securities; delisting returns
Selection bias	Choosing signals, periods, universes after seeing results	Signal library fixed from JKP/OpenAP before modelling; pre-registration; lockbox
Data snooping / multiple testing	Many cells and configurations	Full trial logging; SPA, StepM, MCS; DSR; PBO; haircuts
Overfitting	Complex models	Early stopping; ensembles; validation windows; PBO
Leakage	Overlapping multi-month labels	Purging and embargo equal to horizon
Rebalancing assumptions	Same-close trading	One-day execution lag; drifted weights
Delisting bias	Missing delisting returns	CRSP dlret with convention for missing values
Corporate actions	Price/share adjustments	CRSP adjustment factors; total returns
Point-in-time information	Restatements; macro revisions	Lags; Snapshot robustness; ALFRED or market-based states

5. Validation Approaches: Which Are Appropriate

Approach	Appropriate here?	Reason
Walk-forward with expanding windows	Yes (main)	Mimics real-time research; maximises data for complex models
Rolling windows	Yes (robustness)	Addresses non-stationarity (Capponi et al., 2025)
Nested cross-validation (time-ordered)	Yes, for hyperparameters within the training window	Outer loop = walk-forward; inner loop = validation block
Purged k-fold with embargo	Yes, inside training for early-stopping diagnostics	Required with overlapping labels (Lopez de Prado, 2018)
Standard shuffled k-fold	No	Leaks temporal and cross-sectional dependence
Combinatorial purged CV (CSCV)	Yes, for PBO estimation only	(Bailey et al., 2017)
Single lockbox holdout	Yes (final)	Only credible defence against iterative snooping

6. Reviewer-Proof Checklist (summary)

See Reviewer_Proof_Checklist.md for the full list with owners and evidence locations.

7. Implementation Roadmap (summary)

See Research_Roadmap.md.

Appendix. Experiment Register

Expected outcomes are pre-registered hypotheses, not results. Status tracking in Workbook 4.

ID	Block	Experiment	Hypothesis	Test	Expected (hypothesis)	Priority
E00	Data integrity	Point-in-time panel audit	Panel has no look-ahead: accounting data lagged by availability date; delisting returns applied.	Correlation > 0.95 threshold	Rebuilt factors match library closely	Must
E01	Signal library	Signal replication vs original papers	Rebuilt signals reproduce published long-short t-stats.	Regression; share with $t > 1.96$	Slope near CZ2022's 0.88; R2 high	Must
E02	Signal library	Signal quality and decay profile	Signals differ in IC decay speed and turnover; fast signals are costlier.	Newey-West t on IC	Wide heterogeneity; price-based signals decay fastest	Must
E03	Signal library	Redundancy and clustering	Signals cluster into ~13 themes; within-theme redundancy is high.	Bootstrap cluster stability	Clusters broadly align with JKP themes	Should
E10	Baselines	Naive combinations	Equal-weight z-score, theme-EW, IC-weighted, ICIR-weighted and inverse-vol combinations are strong net-of-cost baselines.	Ledoit-Wolf Sharpe difference	Simple combos have low turnover and competitive net Sharpe	Must
E11	Baselines	Linear statistical combinations	Fama-MacBeth (Lewellen), ridge, elastic net, PLS improve IC over naive.	Clark-West; DM	Modest IC gain; net gain uncertain	Must
E12	Baselines	Mean-variance signal-portfolio combination	KNS-style shrunk SDF over signal portfolios is a strong linear benchmark.	LW2008 test	Competitive gross, higher turnover	Must
E13	Baselines	Published ML benchmarks	GKX-style NN/GBDT and JKMP Portfolio-ML replicate approximately.	Tracking vs published	Directionally consistent replication	Must
E14	Baselines	Linear cost-aware PPP	DMNU-style linear PPP with costs is a strong implementable benchmark.	LW2008; SPA	Beats cost-agnostic linear models net	Must
E20	Factorial core	Cell LS-P: Linear, Static, Prediction loss	Reference cell.	-	Reference	Must
E21	Factorial core	Cell NS-P: Nonlinear, Static, Prediction loss	H1: nonlinearity raises gross IC/Sharpe vs LS-P.	Clark-West; LW2008; SPA	Gross gain; net gain at least 50% smaller	Must
E22	Factorial core	Cell LC-P: Linear, Conditional, Prediction loss	H2a: conditioning adds little net value under prediction loss.	SPA; Clark-West	Small/insignificant net gain	Must
E23	Factorial core	Cell NC-P: Nonlinear, Conditional, Prediction loss	Nonlinear conditioning under prediction loss increases turnover.	SPA; MCS	Higher gross, uncertain net	Must
E24	Factorial core	Cell LS-E: Linear, Static, Economic loss	H3: economic loss improves net utility vs LS-P.	LW2008; SPA	Positive net gain	Must
E25	Factorial core	Cell NS-E: Nonlinear, Static, Economic loss	Nonlinear economic-loss learning (Portfolio-ML-type).	SPA; MCS	Among best static cells	Must
E26	Factorial core	Cell LC-E: Linear, Conditional, Economic loss	H2b: conditioning adds net value when costs are internalised.	SPA; Clark-West-type	Positive net gain (interaction)	Must
E27	Factorial core	Cell NC-E: Nonlinear, Conditional, Economic loss	Full model.	MCS; SPA; DSR; PBO	Best or statistically tied best	Must
E28	Factorial core	Uncertainty dimension (x8 cells)	H4: uncertainty shrinkage reduces turnover and raises net Sharpe.	LW2008; paired bootstrap	Lower turnover; higher net Sharpe	Must
E29	Factorial core	ANOVA-style decomposition of net value	Main effects and interactions of the four factors can be separated.	Block bootstrap CIs	Objective main effect largest; state x objective interaction positive	Must

ID	Block	Experiment	Hypothesis	Test	Expected (hypothesis)	Priority
E30	Portfolio & costs	Implementable efficient frontier	Frontier ranking of cells is stable across AUM.	Bootstrap bands	Cost-aware cells dominate at high AUM	Must
E31	Portfolio & costs	Cost-model sensitivity	Conclusions robust to cost multipliers 0.5x-3x and alternative spread estimators.	Break-even multiplier	Ranking stable up to ~2x [hypothesis]	Must
E32	Portfolio & costs	Capacity	Break-even AUM differs by cell and state.	-	Cost-aware conditional cells have higher capacity	Should
E33	Portfolio & costs	Constraint variants	Long-only benchmark-relative and long-short results agree qualitatively.	LW2008	Qualitatively similar	Should
E34	Portfolio & costs	Heuristic cost mitigation vs optimisation	Optimiser beats buy/hold banding net.	LW2008	Optimiser better at high AUM	Should
E40	Inference	Data-snooping control over all cells	Best cell beats benchmark after accounting for all trials.	Hansen SPA; Romano-Wolf; MCS	Reported honestly whatever the outcome	Must
E41	Inference	Deflated Sharpe and PBO	Selected configuration's Sharpe survives deflation; PBO low.	BLDP2014; BBLZ2017	DSR > 0.95 target [hypothesis]	Must
E42	Inference	Factor-model alphas net of costs	Net alphas vs FF5+UMD, q5 and JKP themes.	Newey-West; GRS-type	Positive alpha vs FF5+UMD [hypothesis]	Must
E43	Inference	Multiverse / non-standard errors	Conclusions hold across reasonable design choices.	Share of specifications with same sign	Main-effect signs stable in >80% of specs [hypothesis]	Must
E50	Robustness	Subperiods	Results hold pre/post-2001 decimalisation, post-2008 and post-2020.	Chow / Bai-Perron	Weaker after 2003 [hypothesis]	Must
E51	Robustness	Universe variants	Results hold ex-microcaps, NYSE-top-500, all stocks.	LW2008	Gains smaller in large caps	Must
E52	Robustness	Horizon and rebalancing	Monthly vs quarterly rebalancing and 1/3/6/12m targets.	-	Longer horizons help net	Should
E53	Robustness	Placebo states	Randomised or shuffled state variables remove conditioning gains.	Permutation test	Placebo gains ~0	Must
E54	Robustness	Pre-publication signal gating	Using only signals published before t does not overturn conclusions.	-	Smaller levels, same ranking [hypothesis]	Must
E55	Robustness	International out-of-sample	Ranking of cells holds in developed ex-US markets.	-	Direction holds with weaker magnitude [hypothesis]	Could
E56	Robustness	Seeds and hyperparameter perturbation	NN results stable across 10+ seeds and +/-1 grid step.	-	Low dispersion relative to cell gaps	Must
E60	Interpretation	Economic feature importance by theme	Net value concentrated in slow themes (value, profitability, investment) and liquidity-aware use of fast themes.	Bootstrap CIs	Value/quality high; short-term reversal low net	Must
E61	Interpretation	State-dependent signal weights	Momentum weight falls in high-volatility post-crash states; short-leg anomaly weight rises with sentiment.	Regression of weights on states	Consistent with DM2016/SYY2012 [hypothesis]	Must
E62	Interpretation	Limits-to-arbitrage mechanism	Gross ML gains concentrate in high-IVOL/illiquid stocks; cost-aware learning shifts exposure away.	Difference tests	Consistent with ACM2023 [hypothesis]	Must
E63	Interpretation	Trading-speed test of CDS2020	Optimal trading speed rises in liquid, persistent states.	Regression	Positive relation [hypothesis]	Should
E64	Interpretation	Alpha-risk alignment diagnostics	Nonlinear alphas show larger ex-post risk underestimation.	-	Larger bias for nonlinear cells	Could

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